

Zaira Anay Espinoza

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PROFESSIONAL SUMMARY

Applied AI and systems engineer experienced in delivering production AI applications from stakeholder discovery and solution architecture through development, evaluation, deployment, and user adoption. Combines expertise in agentic AI, natural language processing, information retrieval, full-stack development, and aerospace systems engineering.

TECHNICAL SKILLS

AI & Machine Learning: Agentic AI, NLP, Text Embeddings, Topic Modeling, BERTopic, LDA, Information Retrieval, Model Evaluation, TensorFlow, PyTorch

Development & Data: Python, SQL, C++, REST APIs, Full-Stack Development, PostgreSQL, SAP HANA, Amazon OpenSearch, Tableau

Cloud & Security: AWS, OpenAI APIs, OAuth, Production Deployment

Systems Engineering: Systems Integration, Verification & Validation, Requirements Analysis, Test Development, Anomaly Resolution

EXPERIENCE

LOCKHEED MARTIN — Engineering Leadership Development Program (ELDP) | Engineer Level 3 | Jan 2024 – Present

- Selected for Lockheed Martin's rotational leadership development program supporting systems engineering, integration & test, enterprise AI solution development, and operational program support.

AI Solutions Engineer — Enterprise Operations

- Lead the end-to-end design and delivery of production AI applications, owning stakeholder discovery, solution architecture, full-stack development, testing, model evaluation, deployment, documentation, user training, and lifecycle support.
- Developed an AI-enabled program-risk analysis solution that reduced a recurring manual process from approximately 16 hours to 25–30 minutes—a roughly 97% reduction—while supporting users across multiple programs and teams.
- Applied text embeddings, BERTopic, and Latent Dirichlet Allocation (LDA) to cluster and classify approximately 3,000 historical program-risk findings, revealing recurring themes used to anticipate risks and inform contract and proposal planning.
- Build agentic AI and information-retrieval capabilities using OpenAI APIs, AWS, and Amazon OpenSearch, enabling applications to retrieve and interpret enterprise information across multi-step workflows.
- Architect secure web applications integrating AI models, APIs, PostgreSQL and SAP HANA data sources, user-facing interfaces, and OAuth-based authentication.

Integration & Test Engineer — Aeronautics

- Performed integration, verification, Final Qualification Testing (FQT), and Failure Mode & Effects Testing (FMET) for classified aerospace subsystems.
- Authored test procedures and executed validation activities within controlled test environments to verify subsystem functionality and software performance.
- Investigated failures and documented anomaly reports detailing technical findings, subsystem impacts, and operational implications for engineering review boards.
- Served as a test SME for aircraft air data system integration and subsystem interactions within highly integrated operational environments.
- Supported onboarding and training of engineers while contributing engineering metrics and program reporting efforts.

LOCKHEED MARTIN AERONAUTICS — Data Analyst | Mar 2022 – Dec 2023

- Supported F-35 digital BOM configuration and serialized component tracking for U.S. and international aircraft deliveries.
- Investigated duplicate or erroneous serial number records for life-limited aircraft components through root-cause analysis across multiple systems and data sources.
- Utilized SQL, scripting, and data analysis techniques to support engineering data validation and process improvement efforts.

EDUCATION

Southern Methodist University — Master of Science in Computer Science, 2025 | GPA: 3.8

Texas A&M University — Bachelor of Science in Industrial & Systems Engineering, Minor in Mathematics, 2021